

ADVENTURES OF FUTURE SCIENCE

April

WONDER Stories

HUGO GERNSBACK
Editor



"50th CENTURY REVOLT"
by Arthur C. Stangland



Other Science Stories
in This Issue

"THE REIGN OF THE
STAR-DEATH"
By A. Rowley Hilliard

"THE ELECTRONIC SIEGE"
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*** START OF THE PROJECT GUTENBERG EBOOK THE
ELECTRONIC SIEGE ***

THE ELECTRONIC SIEGE

By John W. Campbell, Jr.

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JOHN W. CAMPBELL, JR.

Mr. Campbell's humorous story, "The Derelicts of Ganymede," was a huge success. He shows now that he can do more serious work, with just as much skill and power as the humorous efforts. This story has many unusual aspects: it has strange adventures on another world, and a scientific theme that is audacious and yet convincing. There is a mystery to be settled on an inhabited asteroid: strange kidnappings and disappearances in space; a colony of scientific experimenters who defy the rest of humanity.

That is the background of this unusual story, which shows vividly some of the queer tricks that nature might play to upset the best laid plans of mice and men.

The dock-guard looked sourly down the long, bare customs landing. A stream of famous scientists had been coming aboard the space liner *Vega* all afternoon. They were the greatest scientists of Earth and Venus, taking

passage on the chartered liner to Mars for the Interplanetary Society for the Advancement of Science Meeting for 2374, held once each Earth year. Pale faces identified the Venusian colonist-scientists, and the lightly tanned Earth scientists, but the figure coming toward the guard now was obviously neither.

He was a short man, perhaps five feet seven, but he seemed shorter because of the amazing girth he displayed. He must have weighed easily two hundred and seventy-five pounds, his round body mounted on chunky, blocky legs was topped by a neck so thick it was almost missing. He was equipped with a pair of stubby arms on the general plan of the legs. He waddled up on his blocky legs in a most amazing manner, puffing and blowing as ludicrously as an old steam engine of four centuries ago.

But more amazing than his own person was the moving rainbow he represented. Pale green, silken trousers, a deep red shirt, lavender suspenders, and a deep green sash matching his shoes completed his costume. No ancient oriental potentate, even Solomon in all his glory, was not arrayed such as this.

"Must be color-blind to like that," grunted the guard. Then he stared. Behind the man, following on a series of little self-propelled porter trucks, faithfully following the lead-set in the fat man's hand, was a pile of luggage as even this space liner guard seldom saw.

The weighty individual was grunting something to himself as he waddled hastily along, a curious expression of indignation, worry and annoyance intermingled on his puffy-cheeked face.

"Stupidity—rank stupidity of these public officials," he muttered. "Stupidity and red-tape. Will they never realize that it is business that supports them—yes and science too! Refusing Earth's greatest business man passage!

"More officialdom! Pass?" he barked at the dock-guard. "Yes, I've got a pass, and the very devil of a time I had getting it from that idiot, Commander Morrison. Interplanetary Lane Patrol Commander he is—and doesn't recognize the rights of business above science! Yes—Ezra Markham is right. You ought to recognize me—my pictures appear over the visonews frequently enough—and you're a passport guard!"

The grumbling multi-billionaire went toiling up the inclined gangway, the luggage trucks trailing behind like so many pet dogs.

"Steward!" he roared as he stepped off the gangplank.

A steward appeared immediately, dressed in his neat white uniform. He was soon followed by others, rushing to the scene of the emergency the roar seemed to indicate.

"My suite!" demanded Markham, turning over the lead-set and his pass to the handiest one. The startled steward accepted them dumbly, looked down at them, then seemed to remember his duties.

"Yes sir—right this way."

"What—this tiny cubbyhole my room? My closet you mean! Here—show me a suite. Taken? Who—scientists I suppose, scientists that never had brains enough to make a nickel for themselves. Come along—Ah—this will do. Who's in here? Edgar A. Freeman—he's on my engineering staff, I believe. Well—move out his bags and put my cases in. I'm going to the lounge."

The white-coated steward looked in blank surprise at the expansive back of the retreating Mr. Ezra Markham, scratched his head in puzzlement, and went to look up Dr. Edgar A. Freeman, inventor of half a dozen devices which had made space navigation possible on a large scale.

Mr. Ezra Markham seated himself in the largest chair in the main lounge (the only one that would comfortably contain his bulk) and began a very expensive cigar, made of the finest imported Venusian tobacco.

He then spent the next half hour assuring distracted stewards that the rigid "No Smoking" rules didn't apply to Mr. Ezra Markham in the least degree, and in filling the carefully filtered air of the space ship with a variety of noxious odors.

Presently they took off. The slim, graceful ship rose smoothly into the atmosphere, angling slightly up, then driving forward as the gentle hum of mighty Farrel Atomics poured their millions of horsepower into the McKinley space-drive discs.

Markham looked at his jewel-studded wrist-watch and grumbled. They were a few minutes late on the take-off. One of the stewards remarked (under his breath) that that watch must have been made so thick that more jewels might be crowded on.

Shortly after their take-off, a slim, powerfully built man in the white and blue dress uniform of the Interplanetary Lane Patrol came over to Mr. Markham and very definitely pointed out that "No Smoking" meant a complete absence of smoking, whether it be smoke hallowed by the said Mr. Markham, or the second apprentice oiler's assistant.

Mr. Markham put out the stub of his cigar. "I thought," said he testily, "that you men had ships of your own to go in. Since when have Lane Patrolmen been riding passenger liners?"

The Lane Patrolman, not unnaturally proud of the reputation of the Service, replied a bit hotly. "This is a special occasion, as you should know, Mr. Markham. Two years ago a group of scientists calling themselves the Eugenics Research Colony set up an experimental colony on some undiscovered planetoid and have been experimenting on human genetics with captured subjects. Such fine men and women as are aboard this liner would be an obvious prize capture for them, and we are guarding it heavily."

"Humpf. And why has it been going on two years, may I ask?" retorted the disgruntled plutocrat.

"Because there are more than five thousand known asteroids that might house that colony, and because their ships have a weapon we have not been able to duplicate—the protoelectric gun," replied the officer, turning smartly and walking away. Markham chuckled. The Lane Patrolman had retired in poor order.

The E.R.C. was a sore spot on the Lane Patrol's otherwise spotless record. Thanks to a mechanism capable of projecting twin beams of highly dangerous electrons and protons from a "gun", their ships had beaten off patrol cruisers time and again. Secret service men of the Patrol had indeed been able to get into the "Colony"—only to become unwilling subjects to this experiment with human lives. Picked men all, these men of the service, and desirable material for the Research men. Four men so far had been

released after six months' imprisonment of the most heartily detested nature to return with a tale of absolute failure.

Markham rose, and walked back to the aft-walk, a narrow corridor lined on one side with the heavy armour plate of the inner hull, and on the other by six-inch, metal-ribbed unbreakable glass. Here, where only the dimmest lights glowed, the stars and planets shone with the cold, unwinking glory of space, and the sun of space poured down its bluish light in blinding glory.

Behind them now the greater and lesser globes of Earth and Moon were circling in slow majesty, yet from this million mile distance they seemed, and the whole solar system seemed, more as some marvelously ingenious planetarium where unseen supports and unseen wheels drove these visible models of worlds. It seemed impossible they could be real, that Earth with its billions and Moon with its millions of inhabitants were real worlds.

Silently the stocky man watched out into the night. Dimly his eyes made out the dark, torpedo shape of another ship, a Lane Cruiser, some fifty miles astern, and another off to the side. Fifty-man cruisers each one, equipped with the latest devices of offense and defense—but would they be able to drive off the E.R.C. ships?

The dim figure started as he observed the slow blotting out of stars in the Milky Way. Another ship!

Simultaneously there came the soft hum of an alarm signal from within the ship. A moment later the swift tramp of steps, and then a start—a surge of power, and the acceleration that had been held at a comfortable 32 feet per second, Earth normal, was drawn swiftly up to three normal, the highest acceleration a passenger ship could safely stand.

The stocky man, now under a weight of fully eight hundred and twenty-five pounds, should by all rights have lain down at once. Instead his frame seemed to stiffen, the outlines of his body subtly changed, and he glanced along the deserted companion-way hurriedly assuring himself no other was in sight. Then—instead of the infinitely laborious crawl to be expected under nearly half a ton of weight, he walked easily with firm, lithe strides to a sheltered nook behind a great beam. He waited there quietly on his feet! Now there was an amused smile on his face, and the puffiness seemed somehow gone, more an appearance of swelling prevailed. His grey eyes

seemed flashing with interest, his shoulders had lost their slough, and had come back squarely.

The Patrol cruisers were dropping swiftly astern, turning to meet the rapidly approaching sphere ship that was swimming nearer in space. It was maneuvering that the line of the enemy's approach would not bring him directly between them.

Faint halos began to form about the cruisers, and now about the passenger liner. They were the faint halos of magnetic and induction screens that would stop any shell, or any induction ray before it reached the ship. But unfortunately not the twin beams!

Eyes alert, the man watched the sphere come sharply near the two cruisers—a sudden flash, and it was bathed in flaring, shimmering light, warring energy as its powerful screen battled the rays of the Patrol ships.

Two streams of nearly invisible fire reached out in lines as straight as a linear equation toward the two Patrol cruisers. Each was in reality a twin beam, and each was potentially deadly. They snapped out in absolute lines—till they hit the magnetic fields about the patrol ships, then they bent sharply as their electric charges were deflected by the field. Presently they moved swiftly, shifting till the deflection of the field no longer carried them past the ships. Then—a single flash of livid flame, a bursting puff of enveloping fire—and one Patrol ship was punctured, its atmosphere glowing brilliantly under the terrific electronic bombardment.

The Patrol ship, perforce, veered off and soon the second ship came in for a like dose. Again—again they were driven off. It was obvious that the sphere ship did not want to destroy them, merely to cripple them.

Finally each beam split in two, and they wavered and waved weirdly as the ships strove valiantly with their magnetic fields, attempting to divert them sufficiently. Meanwhile a constant terrific bombardment of induction rays was driving at the sphere, its screen flaring vividly.

The two separate beams found their mark, the Patrol ship seemed to glow all over its surface, and instantly every ray, every screen went dead. A

moment later, the second Patrol ship went dead.

The stocky man cursed softly. He turned, and as the ship lurched slightly to a still greater acceleration, nearly three and a half gravities, in its hopeless attempt to flee, he ran lightly down the companion-way, slowing to a walk as he reached the bulkhead door. Then his body seemed to slump, once more his powerful shoulders sagged, and he crawled heavily into the lounge. The other passengers were struggling on hands and knees, or flat on their stomachs, slowly moving along. Finally he fell forward and lay motionless.

Through a great, curving port he could still see the sphere climbing up behind them, rapidly overhauling them. Suddenly two beams of fire were reaching out for their ship—two great beams that bent sharply as they struck the *Vega's* magnetic meteor-shield. Careful manipulation—and Ezra Markham started violently, his muscles contracting, while every sharp bit of metal in the room gave off fans of blue electric fire.

At that same instant every man on the ship fell senseless under the powerful shock transmitted to them. Like the Patrol cruisers, the lifeless, defenseless ship careened on through space, her acceleration dropping immediately to Earth normal as automatic machines took up the unconscious pilot's duties.

A dozen men in silver-grey uniforms were standing about the room when Markham regained consciousness. One of them was bending over him, peering with surprised interest at his magnificent display of coal-tar dyes.

"Who," demanded the E.R.C. man with interest, "are you?"

"Ezra Markham," answered the prostrate man sitting up, "Who are you?"

The lean, tanned face split in a grin. "Tod Morley, not that it makes any difference. And so you're Ezra Markham. Well, well. From the looks of you I should say your type wasn't wanted on our colony. But—it is an experiment for the benefit of the human race, and we feel sure that a man as you would want to contribute to its success—say a million or two. In the meantime—just move over to that corner." A ray pistol in his hand swung on Markham, and the twin rays leaped out. The animate rainbow gave a yell of surprise and pain, and scrambled to his feet with an activity surprising in one so stout. With the inducement of the pistol he was herded hurriedly to the indicated corner.

CHAPTER II

An Artificial World

In the center of the lounge all the scientists were grouped with the captured crew and the I. L. Patrolmen, helplessly hedged in by a barrier of brightly luminous blue-green rays. Professional looking men in simple uniforms were going among them, pointing out one here or there, who was immediately singled out and put in a second, smaller group. Now Markham was marched up, and the entire group efficiently escorted to the communication lock, where a long, flexible tube between the ships permitted them to go from the passenger liner to the space ship without space suits. Twice prisoners rebelled, to be instantly and efficiently paralyzed with a beam from a pistol. In a few seconds the rebellious ones regained consciousness and were able to go on.

The rest of the trip they enjoyed from several small ports. The entire group was penned in a small prison locker with a few guards to watch them. After some six hours of travel, during which the space liner had been left helplessly behind, and the two Patrol cruisers lost in the blackness of space, one of the men in charge came down and gave them a brief lecture on the experiment in which they were soon to take an unwilling part.

It was nearly twenty-four hours later that they finally approached their destination, a tiny, five-mile world of solid metal, a part of the nickel-steel core of some long vanished planet. Its surface turned swiftly beneath them, flashing around in moments as they watched, a surface made up of great crags and clefts of metal, broken, barren masses of metal.

"Lord—it would be impossible to establish a city on the surface of that top!" exclaimed one of the Patrolmen. "The centrifugal spin there would throw anything off into space."

"How about the inside of it then?" asked one of the guards, smiling at him.

"Inside? Is it inside?" asked Markham in alarmed surprise. "Is it always dark?"

The guard laughed and pointed toward a huge cliff of solid metal. "See that—now it's turned by—but that's where we go in. Even from here, what with that fast rotation, you can't see the entrance. See—the ship's going into an orbit about it, picking up to landing speed, trying to move at the same speed as the ground. When the colony was established, the whole interior was carved out with atomic burners—burned the stuff out into gas, and let it escape. The shell's about half a mile thick. Inside, the centrifugal force gives an acceleration just equal to one earth gravity, we're up to speed, and you can see we have about an earth-weight away from it now. And an artificial sun gives plenty of light."

Indeed, as he spoke, they saw the planetoid consistently above them, and now spinning very, very slowly while the heavens gyrated swiftly about. The sun was flashing in and out of the windows in bewildering streaks of light.

Before them the great metal cliff loomed suddenly. Then they saw a huge black gulf under the glare of the spotlights that lighted it during the momentary "nights" of this spinning world.

Into the great tunnel they sank, and the moment they were within it all sense of spin, of rotation, ceased instantly. It seemed they were following some vast tunnel on Earth, where smooth walls of shining metal reflected their spotlights from crystalline metal surfaces.

Far ahead the light was swallowed in blackness. The ship shot straight on, till gradually a curving wall that bent upward appeared, and around the bend the ship sailed. Then a short upward channel and another long level stretch. Light here now, and the spotlights blinked out. The walls were strung with long, thin daylight gas-glow tubes, and far ahead a ring of such tubes surrounded a gigantic metal door, a huge thing that plugged all the great tube.

As they approached, it swung open its sections, lay back along the channel, and closed behind them as the ship came to a halt. Ahead was a similar metal door, and now as the rear door shut there was a terrific shrieking roar as air pounded into the lock, the inner door was opening. Now the tube ahead was ablaze with lights, a short horizontal stretch, then an abrupt rise of several hundred feet—and they had entered their weird artificial world.

In amazed wonder the captives stared out of their ports. Outside the metal ground curved gradually up and away till it arched in a mighty curve far above their heads. The metal here was smooth, obviously fused that the world might be air-tight, free of cracks. Plants, grass, a few trees grew about them, and even above them. And yet there was no more sense of motion here than on Earth, though this world spun far faster.

Directly "above" them was the sun. A huge sphere it was, glowing with the brilliant daylight gas-glow, and beside it was a second great sphere, but glowing with the weird blue-green of mercury vapor.

The rays from this sun were rich in ultra-violet, and they were warming, supplying heat and light to this inner world as does our own sun to the earth.

Streets—metal pavements on which ran swift, stream-lined cars equipped with rubber tires. Planes were unknown here, unmanageable in the tricky air currents and the tricky acceleration effects of centrifugal force. Off to the left of the tunnel entrance was a low, squat building built, as all these buildings of this world, of solid nickel-steel, high grade armour plate. The world itself was a gigantic lump of armour plate.

"Power-house," grunted the patrolman. Its aerials, beam projectors and leads told in an instant what its purpose was. Even now a faint, nearly invisible beam could be seen reaching up through the atmosphere to the artificial sun.

Off to the right, half a mile away, lay a group of buildings; and now the captives were driven toward them. A pair of guards were given the special duty of escorting Mr. Markham to his private room. He was not to be used in the experiments, but held for ransom, and as such was lead to the tower rooms, instead of the regular "quarters" as were the others.

Half an hour later Mr. Ezra Markham was complaining volubly. He had been put in a room, a bare metal room, containing only a table, a chair and a bed.

Could he have a pillow-case? He never could sleep without a pillow-case, he had grown so accustomed to one, and it should be silk. And a rug? The metal floor was so cold on his feet. And hot water to shave with? And did they turn out the sun at night? When was night?

The guard on his floor brought the pillow-case, soon discovered his mistake, and ceased to bring other things. The fat and worried Mr. Markham had an insatiable desire for comfort, and had complained bitterly at the tough, linen pillow-slip.

The guard, in self defense, assured him there was a night, that it was two hours distant, it lasted twelve hours, and that the stars could not be seen through half a mile of metal. But, he added, the room lights of the buildings across the world looked like stars.

Then, quite fatigued, he retired permanently to the guard on the floor below, out of range of Mr. Markham's querulous voice.

True to the promise, night closed in with the extinguishing of the "artificial sun."

Mr. Markham had spent the evening gazing out of his narrow, barred window. He was about fifty feet above the ground, and directly beneath his window was a rose bed, growing in imported soil. Beyond that was a path, and then a broad, soft lawn which stretched out some distance to join finally with the sweeping metal ground of this world.

As the great spheres sputtered into darkness, a slow, firm smile spread across the man's face. He turned from the window. Now again those wide shoulders were firmly back. His head rested squarely on the short, powerful neck held boldly erect, an air of ease and enormous agility in his walk. A lightness, almost restraint was in his step.

His hands were working at his face, and in moments bits of wax were taken from his mouth. He worked swiftly at the washstand a moment removing the last traces of the bothersome face coloring, to reveal a lean, tanned face, almost black from the constant exposure to the unshielded sun of space. Next, cumbersome, heavy shoes were removed and the flapping, gaudy clothes. Finally he stood clothed only in a dark, elastic suit, a pair of soft-soled rubber shoes and the dark-green sash and the lavender suspenders, as well as the wrist-watch.

But now with the floppy garments removed any modern man would have recognized his type instantly. The heavy, blocky legs; the short, squat body, tremendously broad of shoulder and deep of chest; the rather long, powerful arms were all typical of the Jovian colonist. Humans who had spent their lives on that huge, heavy planet developed just such blocky legs and heavy arms—and, more, the tremendous speed of motion and agility the burdensome gravity necessitated. Since Jupiter's gravity was two and a half times Earth's, it was now plain why the acceleration of the ship had not bothered him—scarcely more than polar gravity of Jupiter!

His grey eyes were alight with amusement as he thought of the guard his volubility had driven below. Now Donald Barclay, crack lieutenant of the Interplanetary Lane Patrol was safely barred in a room fifty feet above the metal ground in a solid metal building.

Ready now, he strode lightly to the window, and settled his hands firmly on two of the inch-thick steel bars set in sockets, that barred his window. Silently he began to strain, and great muscle sheaths stood out across his back and shoulders. One bar began to bend in, and the other out, and finally he lifted the two bars from their sockets. A moment later he had two more.

He lay the bars neatly on his bed, and stepped to the window. He stepped across the sill, lowered himself to arm's length outside, then jumped out as far as he could as he let go.

Now the fifty-foot fall would be fairly certain death to most humans. But to a trained Jovian it meant merely a thorough shaking up. Barclay landed heavily but correctly, and in a moment he was on his feet once more, running across the lawn in long, light strides.

But a guard had heard the heavy thump of his landing, and come out to investigate. And despite the dim light from buildings, he saw the shadowy figure running across the lawn and fired at it. The twin fires of his beams immediately revealed him more clearly. A second flash—the man stumbled—then was suddenly leaping in wild, bouncing flight. The guard stood dumbfounded as he saw the man jump about like a rubber ball—twenty feet into the air—thirty feet in a long leap so swift he could not possibly track it

with his beams. The beams were working once more—and the man was running toward him! Now his beam must sweep wider on each of those insanely incredible bounces—

Other guards were coming at his shout—a half dozen of them—and in that instant the shadowy figure was gone!

The others gathered excitedly about him—where—how—what?

None of them even saw the shadow bounce out of a deeper shadow of a rose-bush. He was just suddenly among them, landing in their midst, apparently falling from the other side of the tiny world. And from some other world he was indeed! Half a dozen places at once, bouncing among them, staying among them so they could not beam him, handling the men like children, till in ten seconds they lay stretched out on the metal path and the lawn.

Then Barclay stooped over them an instant, straightened—and was away. But now he carried four of those ray pistols stuck in the deep green sash.

More guards were coming, but they ran to the crumpled bodies. While they strove vainly to wake the men the Jovian had put out, Barclay disappeared in the dimness.

But the alarm was out before he had made half the distance to the power-house—his goal. Would he reach it to hide nearby before the guard there could establish their cordon?

Above the soft thud of his leaping feet he could hear the cries of excited men. Lights were beginning to flash on here and there about the ground of the power-house, shadowy men were running about.

The power-house loomed low, squat before him, a crouching mass against the surrounding darkness. The tiny slits of the windows here and there showed bright streaks, save where the thirty-foot wall about the buildings masked its lower half. The building was enormous—the power plant of a world of science!

Suddenly there was a warning sputter of light about him, like some diffused lightning—the great artificial sun was waking to life as cold tubes were warmed by rapidly warming generators.

Two—one—he was beside the ten-foot thick wall of solid metal that surrounded the power-house. He tensed his muscles, walked back a way limbering once more, then—up-up-up he sailed, his powerful arms outstretched. With a sigh of relief he grasped the edge of the wall and snapped his body over, dropping on the other side to a tree-spotted lawn. Almost simultaneously with his soft landing the great artificial sun came into dim, flickering life. Shouting, calling men were racing across the lawn toward their posts, looking here and there hurriedly—for none would suspect the runaway could have reached this wall and crossed so quickly.

Darting across the lawn, taking advantage of every bit of cover and blessing that winking, sputtering cold light, Don made for the power-house wall.

Then—he was there. His plan was made already. He had seen the enormous steel grille-work door, protected by heavy protoelectric guns and knew he would never be able to enter there. He leaped—his fingers caught in a window-frame twenty feet above him. Then his powerful arms snapped him up against the light, Earth gravity to the next window-slit. Like some weird fly he went up the wall in jumps and spurts. Just shy of the low parapet wall of the roof he listened for a moment. To his ears came mingled, muffled cries—the calls of guards below, the stamp of feet, the dull whirr and hum of huge Farrel atomic generators within the power-house's ten-foot metal walls.

But from the roof—no sound. In an instant he was over, and hidden from the rapidly strengthening light of the artificial sun. He started as a muffled, rapidly increasing thump sounded behind him. A snap, a creak, and a trap-door in the roof opened. Barclay dived suddenly for one of the tiny booths at the side of the broad roof. Inside he found the controls of one of the high-power protoelectric defense guns.

An instant's observation confirmed his belief—these guns were supplied by the great generators in the power-house itself.

"Clever," he muttered, "if they are captured, they can't be turned against the plant, and nothing smaller could phase these walls."

CHAPTER III

In Control

He turned now, and saw a score of men on the roof, rapidly heading for the different booths, armoured like his with four-foot steel walls. He drew two of the protoelectric pistols, adjusted them quickly for minimum power, and swept their brilliant beams quickly over the men. One after another suddenly jerked to rigidity, then slumped helplessly paralyzed. Before the first had reached the ground, the last was falling.

Barclay turned now to the powerful gun he had captured, and quickly spun it on the ship that had brought them—the only machine on this world powerful enough to break open the power-house.

Two enormous brilliant pipes of blue-green fire leaped out and spattered against the ship. Instantly it was wreathed with an aura of shooting flame from the waste energy of the beams. The four-foot armour of the ship was glowing white-hot in a fraction of a second. In a second it was bellying out in gusts of flaming gas; in five seconds a great hole was gaping in the ship's wall, and already the beams were at work on the great generators of the ship.

Taken by surprise, no effective defense was made till the last moment. Then two great beams shot from the ship, still spread as they had been to disable the *Vega*, and they struck the power-house wall. Instantly, as Barclay saw them swing toward him, he released the controls he had held, leaped to the middle of his tiny booth and stood on his toes. As he had expected there came the flaring corona about every sharp point of metal, the numbing, tingling shock as the current raced through the great building as it had raced through the great passenger liner—and every man in that building was helpless! Paralyzed as the passengers and crew of the *Vega* had been.

But Barclay had been as far as possible from any metal object. His feet were encased in heavy rubber shoes, and his contact had been so poor that in the instant the ship's beams operated before, his own locked controls had

driven their beam through the ship's generators, destroying them and extinguishing the rays. Yet he had not fallen.

For a moment he was numb, his muscles responded poorly, and his motions were jerky. But then he was himself again, and in a moment snapped that beam off. The ship was helpless.

He turned, stooped low, and ran along the parapet wall till he could dive across the intervening space to the trap. In it he went, diving down before any beams could reach him.

He caught a handrail, gripped and steadied himself, and came down the stairway in great leaps. Around him was the shrill whine of huge atomic generators, the dull hum of transformers, the whistle of converters, and the steady thunk-thunk-thunk of the huge air-purifiers that supplied this world with air. Before him, massed around the room's high walls, were the great switchboards, a maze of hopelessly confused signs, meters, wiring and giant switches.

One he spotted directly before him, inch-high letters cried out their message of emergency. "Gates."

"Ah—," Barclay grinned. "How helpful." He pulled the switch below the sign. Instantly there was a shrill, rising hum of powerful motors throughout the building. A moment later came the heavy, ringing clangor as huge gates were shut. The motors stopped automatically, the great ten-foot thick metal slabs in place.

He turned his attention now to the men who lay sprawled about on the floor, their arms and legs spread helplessly as the paralyzing electric current caught them. "How kind of that ship," muttered Barclay, catching up one man in each hand and starting for the lockers. "I never expected a break like that." In five minutes he had the entire crew in the locker-room, stripped of weapons, and was ready to give a last look about the room. Then the easily recognized snap and sputter of a beam in action called him. The great main door was being attacked, and already glowed a dull red!

He slammed shut the heavy locker-room door, and ran for the emergency switchboard. He felt sure the outside gun circuits would be controlled here. They were—and just as the inner side of the heavy door began to melt down, the ray was suddenly snapped off as all power was cut from it.

"Whew—that was a bit too close. I'd better make sure of these other things first!" thought Barclay grimly, looking at the heavy door. They'd have to be cut open now—they were welded shut.

Next he investigated the locker-room door, to make certain he wouldn't be attacked. He decided to weld it shut, as the safest way. His hand pistols did that quite quickly. "Now I wonder—they might have a secret, external gate-opening switch.

"If they did—well they won't be able to use it in a little while." Rapidly he dug some metal bars and with his hand pistols welded them solidly across the various gates. He himself could not open them now without first cutting those bars, and that could be done only from the inside.

"Now to give them a little misery!" he smiled as he walked over to the switchboards again. He quickly investigated the various main switchboards, and in ten minutes had unearthed enough information for his purpose. First, the artificial sun was extinguished. Then every power circuit leading out of the plant was cut, save the electromagnetic circuits that held the mass of the artificial sun in place. They wouldn't dare to cut that for power, to do so would bring down the huge spheres of their sun.

Outside he could hear the mutter and roar of the guards over the hum of the generators. From the locker-room came pounding and roars of rage, the mutter of angry voices.

Barclay grinned, and went up a short flight of stairs to the radio room. He took off his lavender suspenders and the green sash, then he unrolled the sash. In it were a number of lengths of a greenish metal tape. From the suspenders came a number of strands of greenish metal wire. He quickly connected the tapes and the wires together in a very definite plan. Finally, from the heavy buckle of the sash, he extracted a tiny atomic generator, capable of generating several hundred horsepower. Next he took off the thick wrist-watch the stewards of the ship had remarked on, and began prying out several of the gems, as well as the face of the watch. Into the gem-sockets he inserted several leads from the atomic generator and from the greenish metal tapes.

At last he was ready. He now had set up, with the aid of the watch and atomic generator, a very serviceable, highly accurate gravitometer, based on the same principle of space fields as the McKinley space-discs. He proceeded to measure the gravitational fields of Jupiter and the sun, and get their angles.

Finally he set down his data on a slip of paper, gathered his apparatus together and sat down at the huge radio set. In a minute and a half he had the complex apparatus tuned correctly, and ready for transmission.

"Lieutenant Donald Barclay I.L.P. calling Commander Morrison," he repeated four times before his answer came. The receiving disc on his set suddenly glowed with flashing colors, and the image of his Commander appeared before him.

"Inside the planetoid G.C. 2759-h according to my gravitometer readings on sun and Jupiter as follows—J-2 53-45, 51 and S-747 -54.49. At present control power-house. Probably not long—come at once and you may be let in. Lock controls here. If necessary break in, the people will all be killed, but you can bring 'em to life again in ninety percent of the cases anyway—I'll take my chance.

"Data on protoelectric gun—," Barclay talked rapidly, telling everything he could about the protoelectric gun which he had disassembled. Morrison would be here with a dozen Patrol ships in three hours now.

Don went down to the power room. He threaded his way through great masses of high-piled apparatus, great hulking bulks that loomed like weird monsters above him, and purred and growled softly to themselves. The lone, tiny figure of the man strode on through the dim-lighted room.

In his ears echoed the eery music of the great generators humming about him, the sullen mutter of the crowd of men outside, and the mirthless, sodden chuckle of the great air apparatus. Barclay shivered—and wondered.

With sudden resolution he strode hastily to the lock controls. He drew down the outer lock control to the open position and watched the meter connected to it. Nothing happened. As he had feared, they had an over-control at the locks themselves. That meant that Morrison would have to blast his way in with the slow, atomic disintegrators, boring a shaft through the solid metal by transmuting the metal to some gas that would escape into space. Perhaps

as much as a foot a minute—but they'd get there. Then all the air would vanish, and in the twinkling of an eye this would be a frozen, solid world, and every human in it would be a solid, frozen corpse. If they froze quickly enough the doctors would bring them to life again later, but if freezing was too slow, permanent death.

The rumbling noises in this reverberating place was working on even Barclay's iron nerves. Finally he returned to the wireless room and listened to the conversation of the various Patrol ships, even called to Morrison to say he could not work the locks.

He crept out on the roof, once each half hour, and looked carefully down. No signs of any disintegrators, they had no mines here, and no portable disintegrators that could eat through the metal walls.

He was sitting quietly, tensely listening to the banter of one patrol to another so he did not see the thing behind him, nor hear the slight swish as a long sharp knife drove through the supposedly solid three-foot wall at his back. The long slim blade came with the faintest rustle, then a slow swish as it cleaved down the length of a man through the armour-plate wall. A blue-clad arm came through, and opened the slit a bit for a face to peer in. Then, with a rush, a man leaped through the paper-thin metal wall and charged across the room. Behind him came another and another!

Barclay was up, and whirled about in an instant, his taut muscles delighting suddenly in the thrill of action against the solid bodies, instead of the phantoms that had been oppressing his waiting mind. The solid thud of fist on flesh, the heave and wrench that tore the long knife from his attacker's arm, the sudden jerk that threw him headlong down the stair. Then he was bouncing about the room like an animated rubber ball, moving with a speed impossible to these slow terrestrials.

But more came—and more! In seconds the room was so crowded he could not dodge, and it was only a question of wrestling, and sheer brute strength—and time. Barclay fought with the strength of five—but there were nearer five score against him. Eventually a stout wooden piece connected with his skull, and he went down.

He came to consciousness in a corner of the room, his wrists and ankles securely bound with wire. Directly before him were the radio-places, with their operators.

"All ships called in for immediate defense against I.L.P. Secret agent escaped in planet and called Patrol. All speed required. Beware Patrol ships on nearing," droned one after another as he picked up one of the nearly two-score E.R.C. ships.

That would not be so good, decided Barclay. He had to do something more.

He turned, looking about for some means of escape. The wire on his wrists was far too strong to break, even though no one was watching him directly.

He could see the wall that had been his downfall now, it had been hollowed out to a mere shell by the work of a disintegrator. He had locked those men in there with a disintegrator! He hadn't had a chance to search the room, and they had simply set it up and cautiously run a hole through the wall, masked by the noise of the generators.

They hadn't given him any disintegrator, that was obvious. And they had very thoroughly searched him now. He looked about at the hard, smooth floor, and the hard, smooth walls, and the equally hard weld that joined them. His eyes lighted in sudden triumph—for that was not smooth! It was merely puddled metal, with the usual slightly rough surface—than which no more perfect file could be imagined!

Barclay started to work, writhing and complaining. "These wires are cutting off circulation—don't have to pull 'em so tight! I even left you a disintegrator, you might at least keep these from cuttin' my wrists off," he grunted. He moved his body about, trying to ease them.

He gave a little louder grunt—to cover the low snap as the weakened wire broke. He had filed it half through (and felt sure his wrist was half filed through too), but now he was able to work on his ankle wires.

It was nearly a quarter of an hour before he had loosened the stiff, tough wire. Cautiously he wriggled his feet into position.

Then, like some mechanical leaping toy, he was on his feet, and before the dozen or so men in the room could move to stop him he was rushing for the stair head. A man leaped in his path—and flew head over heels backward as

a terrific fist caught him on the chest, while a hand like a striking snake's head plucked free his ray pistol.

Barclay leaped—he didn't trouble to use the stair to descend the thirty feet to the floor below. He leaped and darted at once to the great switchboard, running along it while the startled cries of men drowned out the powerful hum of the generators. Running feet behind him—before him—at his side, shouted orders to fire, and cautions to hold fire. He was running in front of irreplaceable control panels, where a ray would mean instant destruction to the planet as the giant powers of the generators would be released explosively.

Men appeared before him like phantoms, and like phantoms disappeared as he seemed to brush them aside. His mind was concentrated on one objective. He found it—pulled a switch. The power-house was suddenly black as every light snapped out.

Instantly his carefully mapped course swung. With a series of leaps he was hidden behind a huge mass of apparatus before the next man could restore the switch, and the daylight tubes came on again.

Barclay was hidden amongst the great mass of apparatus that fed the power to the defense stations on the outer surface of the planet, the huge snaking cables of the protoelectric guns. They had to search for him, and they couldn't use their rays, it was too dangerous. But then, neither could Barclay, so they came in after him. It took ten minutes to dislodge him. The men that came after him were promptly and quietly put out of business for half an hour or so. Finally one man coming around one way attracted Barclay's attention, and he was reaching up for him, when an unseen opponent with a ray pistol brought it down on the back of his head.

CHAPTER IV

Approaching Death!

When he regained consciousness this time he was again in the control room, the radio-places fully occupied, but he was accorded the honor of a private guard.

"Hmm—what are you, guard of honor?" he asked, smiling at the lean, well-built man watching him.

"Cross hairs," affirmed the latter with a smile. "The way you handle your fists deserves honor, we felt. How'd you saw those wires?"

"Do a better welding job next time and it won't happen," laughed Barclay, pointing to the shiny spot on the wall with his head. "How long have I been dreaming?"

"Long enough for the Patrol to come in sight. Morrison has been in communication with Commander Rogers there, and demanded surrender. Rogers invited him in—if he could make it. Our ships haven't showed up yet, but the outside stations are preparing a warm welcome for the Patrol."

Barclay smiled to himself. "Are they? What weapons?"

"You ought to know—protoelectric guns, that's the only power leads you saw wasn't it?"

"Why so centralize your power?" asked the Patrolman.

"Several million dollars cheaper," replied the guard.

"Might have been several lives cheaper not to do so, though."

The guard looked at Barclay sharply. "Why?"

Barclay was starting at the screens on which were depicted the heavens as seen from the outer surface. Stars, planets and the sun were wheeling madly across the sky as the tiny world spun on its axis. On the flashing screens though, Barclay could detect the long, golden form of the Patrol cruisers

coming in massed formation. They spread out now as they neared, circling down at thousands of miles an hour.

"Ray station 21-B when in position will fire. As others come into position join action," came the low firm voice of Commander Rogers of the E.R.G. forces.

"To ships," he continued. "Come with all speed, now attacked by I.L.P."

Suddenly station 21-B, as seen on the screen representing that station, sprang into activity. The great twin barrel of the proto-gun swung on its support, then was suddenly clamped to the gyroscopic rest that would hold it aimed at any one point in the rapidly moving heavens. The patrol cruisers came into view on the eastern horizon—and the gun snapped into action. Instantly one of the barrels glowed with the intense ionization—but the other did not react!

"21-B drawing only negative, sir!" sang out the alarmed voice of the meter-man covering that station. Already the next station was in action, as the Patrol came over its horizon. In moments half a dozen stations were in action—and in every one *but one barrel was working—the negative barrel!*

One short day was over, every station out there was already in operation, and the Patrol ships were already glowing dull red on the outer hull from the impact of fast flying electrons. They retreated hastily to a safer distance.

"21-B reports its apparatus will not shut off!" came the alarmed voice of that station's representative.

"Station 20 reports similar trouble, also drawing only negative!" One after another the reports came in. The men turned in astounded wonder to Commander Rogers. He was looking at Barclay.

"As I said, it might have saved several lives to have spent a few millions for decentralized power. As it is, those stations are projecting electrons away from the planet at an enormous rate, and with tremendous velocity, leaving the planet heavily charged positively. Now—"

"Generator stopping apparatus hopelessly wrecked, sir!" reported an attendant from below!

"—since the generators can't be stopped, and couldn't be started again if they could," continued Barclay, as though explaining some abstruse theory,

"the charge will continue to build up. The stopping apparatus will be completely repaired in half an hour, but the starting voltages and currents will not be available. To stop the generators means the air apparatus would stop, and we'd suffocate in about an hour."

As he spoke, a faint aura was building up about the pointed metallic objects in the room. It was rapidly increasing, spreading fan-wise from every point. Commander Rogers was looking steadily at him.

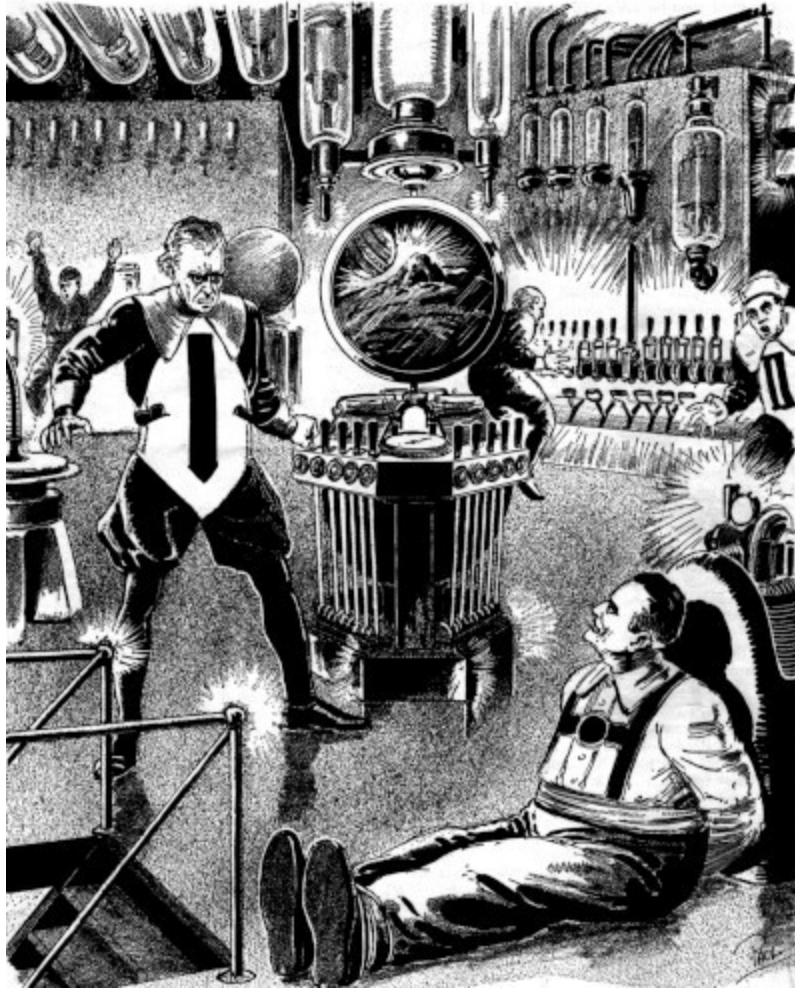
"May I suggest, Commander Roberts, that you warn your ships to avoid the planetoid—they'll be attracted, caught in the electric field, and crushed," continued Barclay. "Also, these bonds are most uncomfortable."

"NO!" roared Rogers. "We won't warn those ships away—they could start the generators for us if they get here—and they will!"

"That's just it, if they get anywhere near—they can't help coming here—much too promptly for their good," replied Barclay seriously. He was enjoying himself.

The patrol cruisers had retreated to a considerable distance. The electric field produced by the tremendously powerful atomic generators was already dangerous, and growing momentarily. Pieces of paper were beginning to act weirdly under the influence of the tremendous fields that already existed. A howling, shrieking wind was building up, and air currents in the room were carrying the impalpable dust into the air, the men began to cough and sneeze. The radio set was functioning poorly, the capacitances tremendously disturbed.

A lightning storm was building up outside the power-house, and terrific, reverberating thunder roared through this miniature iron world, echoing and re-echoing in the closed space. Even the men in the room felt the effects, they were noticeably repelled from the walls, even the floor repelled them. The field was building up with tremendous speed.



A lightning storm was building up outside the power-house, and terrific, reverberating thunder roared through this miniature world.

"And," Barclay reminded Commander Rogers, "the men had better stay away from the generators. Their negative terminals are the only source of electrons on this planet. The voltage will build up to maximum the generators can afford, about 100,000,000 volts isn't it? And then it will remain constant."

Men were already at work on the wrecked stopping apparatus, the huge shunts that could turn the generator coils and their enormous currents into a short-circuit. This would give a polarity reverse kick, halting the disintegration of the atoms. Barclay had fused the apparatus with his pistol.

Already the field growth rate was lessening. The fans of cold fire from every point were lessening, the thunder was gradually diminishing, the generators were working against a greatly increased voltage now.

"Ship 29 reporting—hundred thousand miles to come," reported one of the operators.

"Warn him off—or he'll be wrecked," suggested Barclay.

"Order him on with all speed!" roared Rogers. "And *YOU*—shut up!"

In five minutes the sphere ship was visible as a dot on the screen, rapidly approaching. It seemed to grow with lightning rapidity. "The fool should slow down!" muttered Rogers.

"Ship 29 reports some difficulty—McKinley discs fail to function properly, and he can't slow down."

"If he hits this world, you're going to see a lightning display you never saw the like of before," pointed out Barclay.

The ship was flashing toward them with ever greater speed. It was turning now, struggling to bend its course around the planetoid, but already the great ship was in the clutches of the field. It flashed on, seemed about to skim the surface of the planetoid—

Then they could detect a movement of the entire world! Slight—but enough. The ship collided with a mountain of solid nickel-steel. Instantly there was an enormous blaze of incandescent gas—and then there was neither mountain nor ship—just a spout of gas hurling itself off into space! It shot away from the planetoid as swiftly as the ship had come.

The tiny world seemed to tremble under the concussion of the blow, and then every man felt a sudden swift tingling through him as the electricity stored in his body rushed from it, as it was rushing from his world. And now there broke out the lightning Barclay had prophesied. Terrific stabbing flames reached down out of the atmosphere, snatched at the buildings at every high point. The proto-gun booth outside the window was suddenly wreathed in flame. Then, when the flame was gone, the proto-gun booth with its four-foot steel armour was slumped molten over the wall, like some tired thing. The terrific crashing of the thunder was deafening.

For fully thirty seconds it lasted. Then again it was quiet.

"Better warn the rest of the ships," said Barclay in the hush that followed.

"Ship Fourteen reports powerful electric disturbances from a distance of ten thousand miles."

"Tell ship Fourteen to use no deceleration, and a strong acceleration to one side—she might make an orbit yet. Don't be stubborn, Commander Rogers—there are men on that ship," pleaded Barclay.

"Tell her to circle," said Rogers softly, his voice choking.

The field was increasing no more. Men found their repulsion was so great now that they could not touch the walls of the room, save with considerable effort. With a little jump they could rise feet into the air. It was interesting—but deadly.

"Stopping apparatus repaired, sir," reported a stained mechanic, thrusting his head above the stair-level.

"Asteroid N. G. C. 469 turning in her orbit, approaching us!" called an excited observer. "She's glowing red hot now from the effects of our beams."

"How big?" asked Barclay sharply.

"Quarter of a million tons—metallic," replied his guard.

"Tell Fourteen to lead her—his ship will attract it, and get it in an orbit—if it hits the planet will burst!" Barclay's voice was worried for the first time. "Damn—didn't think of that," he muttered.

"Do it," said Rogers, his voice choked again. "Release the prisoner," he added.

Barclay watched with intense interest. "Call Morrison, ask him to help!" he called. The lone ship was not turning the planetoid enough. "And you might offer to surrender for medical malpractice and use of humans for experimental purpose," he added.

The huge chunk of metal was falling toward them, faster and faster. The lone ship was turning it—gradually, ever so gradually. It could never make

it.

Suddenly the Patrol cruisers saw the danger, and they too added their efforts.

The men watched the screens in tense silence. Would that great mass of solid metal strike—or pass? Already the lightnings were playing again, and their roar and rumble came to them over the panic shouts and cries of fear from the people gathered outside. Guards and captives alike were running madly about, streams of humans converging on the great locks that might let them away from this mad world.

Slowly, slowly the great mass was turning, but swiftly, very very swiftly it was coming nearer.

"She'll pass!" cried one, "She'll skim over that peak there—sure!"

"Can't! That's a quarter of a million tons of steel, and only those tiny ships working at it—but they may be able to bring us back!" retorted another gloomily. Most were silent, just watching that struggle of tiny ships and mighty masses and terrific electric fields.

Then—it was over! The enormous boulder of metal skimmed less than fifty feet above a high-rising pinnacle of steel, so close that but for the absolute lack of atmosphere, there must surely have been a flaring arc that would have carried all to death.

The lightnings were fading now, their rumbling thunders diminishing.

"That's past now, Commander Rogers—but others, and others will come. Commander Morrison has your twin-ray guns, and he can charge his ship similarly, and approach your entrance there without danger—it won't take long to remove the field then."

Commander Rogers seemed to slump, then he turned to the radio men. "Tell Morrison," he ordered in a low voice, "his men are too good for us. We will surrender to a charge of medical malpractice and use of humans for experiments. We have taken no lives."

THE END

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